

Advance Java Solution-1

1. Write a program in java to demostrat concept of thread.

```
class aa extends Thread{
    public void run(){
        System.out.print("java");
    }
}
class bb extends Thread{
    public void run(){
        System.out.print("Thread");
    }
}
class thread2{
    public static void main (String args[]){
        aa obj1 = new aa();
        bb obj2 = new bb();
        obj1.start();
        obj2.start();
    }
}
```

2. Write a program to print tables of 2,5 and 10 using multi threading.

```
class aa extends Thread{
    public void run(){
        int n=2;
        for(int i=1; i<=10; i++){
            System.out.println(+n+" * "+i+" = "+(n*i));
        }
    }
}
class bb extends Thread{
    public void run(){
        int n=5;
        for(int i=1; i<=10; i++){
            System.out.println(+n+" * "+i+" = "+(n*i));
        }
    }
}
class cc extends Thread{
    public void run(){
        int n=10;
```

```

        for(int i=1; i<=10; i++){
            System.out.println(+n+" * "+i+" = "+(n*i));
        }
    }
}
class table_with_thread{
    public static void main (String args[]){
        aa obj1 = new aa();
        bb obj2 = new bb();
        cc obj3 = new cc();
        obj1.start();
        obj2.start();
        obj3.start();
    }
}

```

3. Write a program to print tables of 2,5 and 10 using multi threading and synchronization.

```

class Table{
    synchronized void printTable(int n){
        for(int i=1; i<=5; i++){
            System.out.println(n+" * "+i+" * "+(i*n));
            try{
                Thread.sleep(1000);
            }
            catch(Exception e){ }
        }
    }
}
class MyThread1 extends Thread{
    Table t;
    MyThread1(Table t){
        this.t=t;
    }
    public void run(){
        t.printTable(2);
    }
}
class MyThread2 extends Thread{
    Table t;
    MyThread2(Table t){
        this.t=t;
    }
    public void run(){
        t.printTable(5);
    }
}

```

```

    }
}
class MyThread3 extends Thread{
    Table t;
    MyThread3(Table t){
        this.t=t;
    }
    public void run(){
        t.printTable(10);
    }
}
public class syn_with_table{
    public static void main(String args[]){
        Table obj=new Table();

        MyThread1 t1=new MyThread1(obj);
        MyThread2 t2=new MyThread2(obj);
        MyThread3 t3=new MyThread3(obj);

        t1.start();
        t2.start();
        t3.start();
    }
}

```

4. Write a Java program to design student regiter page using ATW controls.

Fields : Name, Address, Gender, City, Mobile Number.

```

import java.awt.*;
import java.awt.event.*;

public class s_register extends Frame{
    public s_register(){
        //set the frame
        setTitle("Student Registration Form");
        setSize(400,600);
        setLayout(null);
        setVisible(true);

        //label name and textfield
        Label lbl_sn=new Label("Student Name : ");
        lbl_sn.setBounds(50,100,100,30);
        add(lbl_sn);
        TextField txtsn=new TextField();
        txtsn.setBounds(150,100,200,30);
    }
}

```

```
add(txtsn);
```

```
//label mobile number and textfield
```

```
Label lbl_mn=new Label("Mobile Number : ");
```

```
lbl_mn.setBounds(50,150,100,30);
```

```
add(lbl_mn);
```

```
TextField txtmn=new TextField();
```

```
txtmn.setBounds(150,150,200,30);
```

```
add(txtmn);
```

```
//radio Button for Gender
```

```
Label gen=new Label("Gender : ");
```

```
gen.setBounds(50,200,100,30);
```

```
add(gen);
```

```
CheckboxGroup cbox=new CheckboxGroup();
```

```
Checkbox male=new Checkbox("Male",cbox,true);
```

```
male.setBounds(150,200,60,30);
```

```
add(male);
```

```
Checkbox female=new Checkbox("female",cbox,false);
```

```
female.setBounds(220,200,80,30);
```

```
add(female);
```

```
//Dropdown for Course Selection
```

```
Label con=new Label("City : ");
```

```
con.setBounds(50,250,100,30);
```

```
add(con);
```

```
Choice cou_ch=new Choice();
```

```
cou_ch.add("Select City");
```

```
cou_ch.add("Bhavnagar");
```

```
cou_ch.add("Ahemdabad");
```

```
cou_ch.add("Rajkot");
```

```
cou_ch.setBounds(150,250,200,30);
```

```
add(cou_ch);
```

```
//label address and textarea
```

```
Label lbl_add=new Label("Address : ");
```

```
lbl_add.setBounds(50,300,100,30);
```

```
add(lbl_add);
```

```
TextArea t_add=new TextArea();
```

```
t_add.setBounds(150,300,200,80);
```

```
add(t_add);
```

```
//label password and textfield
```

```
Label lbl_ps=new Label("password : ");
```

```
lbl_ps.setBounds(50,400,100,30);
```

```
add(lbl_ps);
```

```

TextField txtps=new TextField();
txtps.setBounds(150,400,200,30);
add(txtps);

//Button
Button btn1=new Button("Register");
btn1.setBounds(100,450,80,30);
add(btn1);
Button btn2=new Button("Clear");
btn2.setBounds(200,450,80,30);
add(btn2);

//window close event
addWindowListener(new WindowAdapter(){
    public void windowClosing(WindowEvent e){
        dispose();//System.exit(0);
    }
});
}
public static void main(String args[]){
    new s_register();
}
}

```

5. Write a Java program to display employee salary page. (Take fields as you need).

```

import java.awt.*;
import java.awt.event.*;

public class salarypage extends Frame {
    Label l1, l2, l3, l4, l5;
    TextField t1, t2, t3, t4, t5;
    Button b1;

    salarypage() {
        setLayout(new GridLayout(6, 2, 10, 10));

        l1 = new Label("Employee id:");
        t1 = new TextField();

        l2 = new Label("Employee name:");
        t2 = new TextField();

        l3 = new Label("Basic salary:");
        t3 = new TextField();
    }
}

```

```

l4 = new Label("Allowances:");
t4 = new TextField();

l5 = new Label("Eductions:");
t5 = new TextField();

b1 = new Button("Calculate Net Salary");

add(l1); add(t1);
add(l2); add(t2);
add(l3); add(t3);
add(l4); add(t4);
add(l5); add(t5);
add(b1);

setTitle("Salary Details");
setSize(400, 300);
setVisible(true);

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent e) {
        dispose();
    }
});
}
public static void main(String[] args) {
    new salarypage();
}
}

```

6. Write a java program to create invoice page.

Fields : Product No, Product Name, Price, Qty, Discount, SGST, CGST, Total Amount.

```

import java.awt.*;
import java.awt.event.*;

class invoice extends Frame implements ActionListener{
    Label lbl_pn, lbl_pno, lbl_qty, lbl_pri, lbl_sgst, lbl_cgst, lbl_dis, lbl_t;
    TextField txtpn, txtpno, txtqty, txtpri, txtsgst, txtcgst, txtdis, txtt;
    Button btn1, btn2;

    invoice(){
        //set the frame
        setTitle("Invoice Page");
    }
}

```



```
setSize(400,600);
setLayout(null);
setVisible(true);

//label Product Name and textfield
lbl_pn=new Label("Product Name : ");
lbl_pn.setBounds(50,100,100,30);
add(lbl_pn);
txtpn=new TextField();
txtpn.setBounds(150,100,200,30);
add(txtpn);

//label Product number and textfield
lbl_pno=new Label("Product No : ");
lbl_pno.setBounds(50,150,100,30);
add(lbl_pno);
txtpno=new TextField();
txtpno.setBounds(150,150,200,30);
add(txtpno);

//label Quantity and textfield
lbl_qty=new Label("Product Quantity : ");
lbl_qty.setBounds(50,200,100,30);
add(lbl_qty);
txtqty=new TextField();
txtqty.setBounds(150,200,200,30);
add(txtqty);

//label Price and textfield
lbl_pri=new Label("Product Price : ");
lbl_pri.setBounds(50,250,100,30);
add(lbl_pri);
txtpri=new TextField();
txtpri.setBounds(150,250,200,30);
add(txtpri);

//label sgst & cgst and textfield
lbl_sgst=new Label("SGST : ");
lbl_sgst.setBounds(100,300,50,30);
add(lbl_sgst);
txtsgst=new TextField();
txtsgst.setBounds(150,300,50,30);
add(txtsgst);

lbl_cgst=new Label("CGST : ");
lbl_cgst.setBounds(250,300,50,30);
```

```

add(lbl_cgst);
txtcgst=new TextField();
txtcgst.setBounds(300,300,50,30);
add(txtcgst);

//label Discount and textfield
lbl_dis=new Label("Product Discount : ");
lbl_dis.setBounds(50,350,100,30);
add(lbl_dis);
txtdis=new TextField();
txtdis.setBounds(150,350,200,30);
add(txtdis);

//label Total and textfield
lbl_t=new Label("Total : ");
lbl_t.setBounds(50,400,100,30);
add(lbl_t);
txtt=new TextField();
txtt.setBounds(150,400,200,30);
add(txtt);
//Readonly
txtt.setEditable(false);

//Button
btn1=new Button("Total");
btn1.setBounds(100,450,80,30);
add(btn1);
btn2=new Button("Clear");
btn2.setBounds(200,450,80,30);
add(btn2);
//Button action(action performed auto call)
btn1.addActionListener(this);

//window close event
addWindowListener(new WindowAdapter(){
    public void windowClosing(WindowEvent e){
        dispose();//System.exit(0);
    }
});
}
public void actionPerformed(ActionEvent e){
    int qty = Integer.parseInt(txtqty.getText());
    int pri = Integer.parseInt(txtpri.getText());
    float sgst = Float.parseFloat(txtsgst.getText());
    float cgst = Float.parseFloat(txtcgst.getText());
    int dis = Integer.parseInt(txtdis.getText());

```



```

float baseAmount = pri * qty;
float sgstAmount = (baseAmount * sgst) / 100;
float cgstAmount = (baseAmount * cgst) / 100;

float total = baseAmount + (sgstAmount + cgstAmount) - dis;
txtt.setText(String.valueOf(total));
}
public static void main(String[] args){
    new invoice();
}
}

```

7. Write a Java program to take 2 numbers and display its sum.

```

import java.awt.*;
import java.awt.event.*;

class sum extends Frame implements ActionListener{
    Label l1,l2,l3;
    TextField t1,t2,t3;
    Button b;
    sum(){
        //Label
        l1=new Label("Enter First Number : " );
        l2=new Label("Enter Second Number : " );
        l3=new Label("Total : " );

        //TextField
        t1=new TextField();
        t2=new TextField();
        t3=new TextField();

        //Readonly
        t3.setEditable(false);

        //Button
        b=new Button("calculate");
        setLayout(new GridLayout(4,2,10,10));

        //Add components
        add(l1);  add(t1);
        add(l2);  add(t2);
        add(l3);  add(t3);
    }
}

```

```

add(b);
//Button action(action performed auto call)
b.addActionListener(this);

//Frame setting
setTitle("Addition using AWT");
setSize(500,500);
setVisible(true);

//close window event
addWindowListener(new WindowAdapter(){
    public void windowClosing(WindowEvent e){
        dispose();
    }
});
}
//end constructor ction performed
public void actionPerformed(ActionEvent e){
    int n1=Integer.parseInt(t1.getText());
    int n2=Integer.parseInt(t2.getText());
    int sum=n1+n2;
    t3.setText(String.Valueof(sum));
}
public static void main(String[] args){
    new sum();    //constructor call
}
}

```

8. Write a Java program to take 3 numbers and display maximum number.

```

import java.awt.*;
import java.awt.event.*;

class max_n extends Frame implements ActionListener {
    TextField num1Field, num2Field, num3Field, resultField;
    Button maxButton;

    public max_n() {
        setLayout(new FlowLayout());

        add(new Label("Number 1 :"));
        num1Field = new TextField(10);
        add(num1Field);

        add(new Label("Number 2 :"));

```

```

num2Field = new TextField(10);
add(num2Field);

add(new Label("Number 3 :"));
num3Field = new TextField(10);
add(num3Field);

maxButton = new Button("Find Max");
maxButton.addActionListener(this);
add(maxButton);

add(new Label("Maximum : "));
resultField = new TextField(10);
resultField.setEditable(false);
add(resultField);

setTitle("Find Maximum Number");
setSize(250, 180);
setVisible(true);

addWindowListener(new WindowAdapter() {
    public void windowClosing(WindowEvent we) {
        dispose();
    }
});
}

public void actionPerformed(ActionEvent e) {
    try {
        int num1 = Integer.parseInt(num1Field.getText());
        int num2 = Integer.parseInt(num2Field.getText());
        int num3 = Integer.parseInt(num3Field.getText());

        int max = num1;
        if (num2 > max) max = num2;
        if (num3 > max) max = num3;

        resultField.setText(String.valueOf(max));
    } catch (NumberFormatException ex) {
        resultField.setText("Invalid input");
    }
}

public static void main(String[] args) {
    new max_n();
}
}

```

9. Write a Java program to take 3 numbers and display minimum number.

```
import java.awt.*;
import java.awt.event.*;

class min_n extends Frame implements ActionListener {
    TextField num1Field, num2Field, num3Field, resultField;
    Button minButton;

    public min_n() {
        setLayout(new FlowLayout());

        add(new Label("Number 1 :"));
        num1Field = new TextField(10);
        add(num1Field);

        add(new Label("Number 2 :"));
        num2Field = new TextField(10);
        add(num2Field);

        add(new Label("Number 3 :"));
        num3Field = new TextField(10);
        add(num3Field);

        minButton = new Button("Find Min");
        minButton.addActionListener(this);
        add(minButton);

        add(new Label("Minimum : "));
        resultField = new TextField(10);
        resultField.setEditable(false);
        add(resultField);

        setTitle("Find Minimum Number");
        setSize(250, 180);
        setVisible(true);

        addWindowListener(new WindowAdapter() {
            public void windowClosing(WindowEvent we) {
                dispose();
            }
        });
    }
}
```

```
public void actionPerformed(ActionEvent e) {
    try {
        int num1 = Integer.parseInt(num1Field.getText());
        int num2 = Integer.parseInt(num2Field.getText());
        int num3 = Integer.parseInt(num3Field.getText());

        int min = num1;
        if (num2 < min) min = num2;
        if (num3 < min) min = num3;

        resultField.setText(String.valueOf(min));
    } catch (NumberFormatException ex) {
        resultField.setText("Invalid input");
    }
}

public static void main(String[] args) {
    new min_n();
}
}
```